25901126 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBOTA05 - BIO-INSTRUMENTATION AND QUALITY ANALYSIS 2

(2024 ADMISSION ONWARDS)

Duration: 2 Hour **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One or Two Sentences

Answer any **10** questions

Each question carries **2** marks

1. What are the commonly used detergents for cell disruption? [K] / [CO1]
2. Name the mobile phase and one type of stationary phase used in Gas Chromatography [U] / [CO2]
3. Explain gel casting tray. [U] / [CO3]
4. What is the role of fluorophore conjugated antibodies in immunophenotyping? [U] / [CO4]
5. List any two applications of the filtration technique. [K] / [CO1]
6. Mention one advantage of high-performance thin-layer chromatography (HPTLC) over conventional TLC. [A] / [CO2]
7. What are the monomers used in preparing gel during PAGE? [U] / [CO3]
8. What is forward scatter (FSC) in flow cytometry? [K] / [CO4]
9. Explain why osmotic shock is effective mainly for Gram-negative bacteria? [K] / [CO1]
10. List two applications of cell counting in cell culture. [U] / [CO4]

11. What is the main purpose of centrifugation in laboratories? [K] / [CO1]

12. List two types of camera lucida. [U] / [CO4]

[2x10 = 20]

Part B

Short Answer

Answer any **8** questions

Each question carries **5** marks

13. Write down the principle and procedure associated with conventional filtration using filter paper. [K] / [CO1]

14. Explain the various detection methods used in chromatography. [U] / [CO2]

15. Explain the procedure of Western blotting. [U] / [CO3]

16. Explain the principle and application of micrometers in microscopy? [U] / [CO4]

17. Write a note on the refrigerated centrifuge. [K] / [CO1]

18. With diagram explain the principle and procedure of anion exchange chromatography. [U] / [CO2]

19. Write a note on the gel documentation system associated with electrophoresis. [K] / [CO3]

20. Explain the application of cell counting in hematology. [A] / [CO4]

21. What is differential and density gradient centrifugation? [K] / [CO1]

22. Explain the concept and applications of single cell analysis. [U] / [CO4]

[5x8 = 40]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

23. Discussion in detail the principle, instrumentation and working of HPLC. Compare normal phase and reverse phase HPLC. [U] / [CO2]

24. Bring out the applications, advantages and limitations of 2 D Electrophoresis. [U] / [CO3]

[10x1 = 10]